

Distinguishing between the effects of ocean acidification and ocean carbonation in the coccolithophore *Emiliania huxleyi*

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ABSTRACT: The coccolithophore *Emiliania huxleyi* was cultured under a broad range of carbonate chemistry conditions to distinguish the effects of individual carbonate system parameters on growth, primary production, and calcification. In the first experiment, alkalinity was kept constant and the fugacity of CO₂ (fCO₂) varied from 2 to 600 Pa (1 Pa ≈ 10 μatm). In the second experiment, pH was kept constant (pH_{free} = 8) with fCO₂ varying from 4 to 370 Pa. Results of the constant-alkalinity approach revealed physiological optima for growth, calcification, and organic carbon production at fCO₂ values of ~ 20 Pa, ~ 40 Pa, and ~ 80 Pa, respectively. Comparing this with the constant-pH approach showed that growth and organic carbon production increased similarly from low to intermediate CO₂ levels but started to diverge towards higher CO₂ levels. In the high CO₂ range, growth rates and organic carbon production decreased steadily with declining pH at constant alkalinity while remaining consistently higher at constant pH. This suggests that growth and organic carbon production rates are directly related to CO₂ at low (sub-saturating) concentrations, whereas towards higher CO₂ levels they are adversely affected by the associated decrease in pH. A pH dependence at high fCO₂ is also indicated for calcification rates, while the key carbonate system parameter determining calcification at low fCO₂ remains unclear. These results imply that key metabolic processes in coccolithophores have their optima at different carbonate chemistry conditions and are influenced by different parameters of the carbonate system at both sides of the optimum.

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